Familiarization with relevant international conventions and with IHO. national and relevant international codes .

. 2 Ship design and equipment of chemical tankers

A brief description of specialized piping . pumping and tank arrangements, overflow control. Types of cargo pumps and their application to various types of cargo. Tank c le aning and gas-freeing systems. Cargo tan k venting and accommodation ventilation, airlocks. Gauging system s and alarms . Tank temperature control systems. The safet y fa cto rs of e l ectrical system s .

3 Cargo characteristics

Sufficient knowledge o f liquid chemical ca rgo c haracteristics t o enable the proper use of relevant in ternation a l codes .

4 Ship operatioDs

Cargo calculations. Loading an d di scharging p l ans. Loading and discharge procedure. Check lists. Use of moni tor ing equ i pment. Gas-freeing operations and tank c leaning operations (proper use of

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SOURCE: MSC.33(63).pdf

4 Ship operatioDs

Cargo calculations. Loading an d di scharging p l ans. Loading and discharge procedure. Check lists. Use of moni tor ing equ i pment. Gas-freeing operations and tank c leaning operations (proper use of absorption and wetting agents and detergent s) . Us e and maint enan ce of inert atmospheres. Control of entry into pump - rooms and enclosed spaces. Use of detecting and safety equipment. Disposal of waste and washings

5 Repair and maintenance

Precautions to be taken before the r e pair and maintenance of pumping, piping, electrical and control syst ems .

. 6 z.ergency operations

The importance of developing ships' emerg ency plan s. Cargo operati ons emergency shutdown. Action i n the eve n t o f failure of services essential to cargo. fire -fi ghting on c hemi cal t ankers. Action following collision, stranding or spillage. fir st aid procedure and the use of r esus ci tati on and deco nt amination equipment. Use of breathing apparatu s. Res c ue f rom enclosed spaces.

Appendix 4 to regula tion VI I

Liquefied gas tankers

The spe cialized training programme referred to in par agraph 2.2 of

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SOURCE: MSC.33(63).pdf

equipment. Use of breathing apparatu s. Res c ue f rom enclosed spaces.

Appendix 4 to regula tion VI I

Liquefied gas tankers

The spe cialized training programme referred to in par agraph 2.2 of regulation VII appropriate to the duties on liquefied gas tankers shall provide theoretical and practical knowledge of the following sub j ec ts:

.1 Regulations and codes of practice
familiariza t ion with relevant i nt e rnati ona l conventions and with
IMO. national and relevant. i nter nati ona l codes .
RESOLUTION MSC.33(63)
adopted on 23 May 1994
ADOPTION OF AMENDMENTS TO THE INTERNATIONAL CONVENTION
ON STANDARDS OF TRAINING, CERTIFICATION AND
WATCHKEEPING FOR SEAFARERS, 1978 _ 7 _
.2 Ship design and equipment of liquefied gas tankers, incuding
familiarization with:
.2.1 Types of liquefied gas tankers;
.2.2 Cargo containment systems (construction, surveys);
.2.3 Cargo-handling equipment (pumps, piping syste ms) ;
.2.4 Cargo conditioning systems (warm-up, cool-down);
.2 . 5 Tank atmosphere control systems (inert g8S, nitrogen);
.2.6 Instrumen tation of cargo containment and handling systems;
.2.7 Fire-fighting systems; and
.2.8 Safety and rescue equipment .

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SOURCE: MSC.33(63).pdf
.2 . 5 Tank atmosphere control systems (inert g8S, nitrogen);
.2.6 Instrumen tation of cargo containment and handling systems;
.2.7 Fire-fighting systems; and
.2.8 Safety and rescue equipment .
. 3 Fire - fighting
Advanced practical fire-fighting techniques and tactics applicable
to gas tankers, including the use of water-spray systems .
4 Chemistry and physics
An introduction to basic chemistry and physics as it relates to the
safe carriage of liquefied gases in bulk in ships:
.4.1 Properties and characteristics of liquefied gases and their
vapours:
.4.1.1
.4.1.2
.4.1.3
.4.1.4
.4.1.5
.4.1.6
.4 .1 .7
.4.1.8
.4. 1.9
.4.1.10
.4.1.11
.4 .1.12
.4.1.13
.4.1.14
. 4.1.15
Definition of g8S;
Simple gas laws:
Gas equation;
Density of gases;
Diffusion and mixing in gases;

Compression of gases;
Liquefaction of gases;
Refrigeration of gases:
Critical temperature;
Practical significance of flashpoint;
Upper and lower explosive limits;
Auto-ignition temperature;
Compatibility of gases;
Reactivity; and
Polymerization. inhibitors .
.4.2 Properties of single liquidS
.4.2.1
.4.2.2
.4.2.3
.4.2.4
4.2.5
Densities of liquids and vapours;

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SOURCE: MSC.33(63).pdf
Upper and lower explosive limits;
Auto-ignition temperature;
Compatibility of gases;
Reactivity; and
Polymerization. inhibitors .
.4.2 Properties of single liquidS
.4.2.1
.4.2.2
.4.2.3
.4.2.4
4.2.5
Densities of liquids and vapours;
Variation with temperature;
Vapour pressure and temperature;
Entalphy; and
Vaporization and boiling liq uids .
.4.3 Nature and properties of solutions
.4.3.1
.4.3.2
.4.3.3
Solubility of gases in liquids;
Miscibility between liquids and effects of
temperature change ;
Densities of solutions and dependen ce on temperature
and concentration;
RESOLUTION MSC.33(63)
adopted on 23 May 1994
ADOPTION OF AMENDMENTS TO THE INTERNATIONAL CONVENTION
ON STANDARDS OF TRAINING, CERTIFICATION AND
WATCHKEEPING FOR SEAFARERS, 1978 .4.3.4.
.4.3.5
.4.3.6
.4.3.7

.4.3.8
. 4.3.9
- 8 -
Effects of dissolved substances on melting and
boiling points:
Hydrates, formation and dispersion;
Hygroscopicit.y;
Drying of air and other gases;
Dew point; and
Low temperature effects .
.5 Health hazards
.5.1
.5.2
. 5 .3
Toxicity
.5.1.1
.5.1.2
.5.1.3
5.1.4
Hodes by which lique fi ed gases and their vapours may
be toxic;

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